



Safety Data Sheet

Page 1 of 8

LOCTITE 454

Item No.649680

V001.0

Revision: 03.06.2022

printing date: 22.05.2023

MSDS No:AA00514-0000649680

1. Identification of the substance/preparation and of the company/undertaking

A. Product name: LOCTITE 454

B. Purpose of the product and limitations:

Purpose of the product Adhesive
Limitations Prohibition on use except the above

C. Supplier information

Importer: Henkel Korea Ltd. - Seoul Branch, 04177, 1st Floor, Henkel Tower Bldg., 41, Mapo-daero 4da-gil, Mapo-gu, Seoul, Korea. Phone : +82-2-3279-1700

Emergency information: +82-2-3279-1700 or Emergency tel : +82-2-3279-1707 (24hr)

D. Writer division and contact person:

Product Safety & Regulatory Affairs for South Korea, msdsakorea@henkel.com

2. Hazards and Risk identification

A. Hazard Classification:

<u>Hazard Class</u>	<u>Hazard Category</u>	<u>Target organ</u>
Flammable liquids	Category 4	
Skin corrosion/irritation	Category 2	
Serious eye damage/eye irritation	Category 2	
Target Organ Systemic Toxicant - Single exposure	Category 3	respiratory tract irritation

B. Item of labeling with Risk and Safety assessment phrase:

Hazard pictogram:



Signal word:

Warning

Hazard statement:	H227 Combustible liquid. H315 Causes skin irritation. H319 Causes serious eye irritation. H335 May cause respiratory irritation.
Precautionary Statement(s):	
Prevention:	P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. P261 Avoid breathing mist/vapours. P264 Wash thoroughly after handling. P271 Use only outdoors or in a well-ventilated area. P280 Wear protective gloves/protective clothing/eye protection/face protection.
Response:	P302+P352 IF ON SKIN: Wash with plenty of water. P304+P340+P312 IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or physician if you feel unwell. P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. P321 Specific treatment (see on this label). P332+P313 If skin irritation occurs: Get medical advice/attention. P337+P313 If eye irritation persists: Get medical advice/attention. P362+P364 Take off contaminated clothing and wash it before reuse. P370 + P378 In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.
Storage:	P403+P233 Store in a well-ventilated place. Keep container tightly closed. P405 Store locked up.
Disposal:	P501 Dispose of contents/container in accordance with national regulation.

C. Possible Hazards: None if used properly.

3. Composition / information on ingredients

General description: Mixture

Chemical name	Other name(s)	CAS No. or ID No.	Contents (%)
Ethyl 2-cyanoacrylate	2-Propenoic acid, 2-cyano-, ethyl ester	7085-85-0	>= 85 - < 90 %

All remaining chemical compositions are below threshold limit and are not subjected to GHS classification according to the Ministry of Employment and Labor Public Notice.

4. First aid measures

A. After eye contact: If the eye is bonded closed, release eyelashes with warm water by covering with wet pad. Keep eye covered until debonding is complete, usually within 1-3 days. Cyanoacrylate will bond to eye protein and will cause periods of weeping which will help to debond the adhesive. Do not force eye open. Medical advice should be sought in case solid particles of cyanoacrylate trapped behind the eyelid cause any abrasive damage.

- B. After skin contact:** If lips are accidentally stuck together apply warm water to the lips and encourage maximum wetting and pressure from saliva inside the mouth. Peel or roll lips apart. Do not try to pull the lips apart with direct opposing action. Cyanoacrylates give off heat on solidification. In rare cases a large drop will generate enough heat to cause a burn. Burns should be treated normally after the adhesive has been removed from the skin. Do not pull bonded skin apart. It may be gently peeled apart using a blunt object such as a spoon, preferably after soaking in warm soapy water.
- C. After inhalation:** Move to fresh air. If symptoms persist, seek medical advice.
- D. After ingestion:** Ensure that breathing passages are not obstructed. The product will polymerise immediately in the mouth making it almost impossible to swallow. Saliva will slowly separate the solidified product from the mouth (several hours).
- E. Others / Medical advice:** Treat symptomatically and supportively.

5. Explosion / Fire fighting measures

- A. Suitable (and unsuitable) extinguishing media:**
Suitable extinguishing media: Extinguish using agent suitable for type of surrounding fire.
- B. Special exposure hazard arising from product itself:**
Hazardous combustion products: Toxic and irritating vapors.
- Fire and Explosion Risk:** In the event of a fire, carbon monoxide (CO) and carbon dioxide (CO₂) can be released.
- C. Special protective equipments for firefighters and safety measures:**
Wear self-contained breathing apparatus.

6. Accidental release measures

- A. Personal precautions / measures and equipments:**
Ensure adequate ventilation.
Avoid contact with skin and eyes.
Wear protective equipment.
- B. Environmental precautions / measures:**
Do not empty into drains / surface water / ground water.
- C. Methods of cleaning up / removing:**
Do not use cloths for mopping up. Flood with water to complete polymerization and scrape off the floor. Cured material can be disposed of as non-hazardous waste. Dispose of contaminated material as waste according to Section 13.

7. Handling and storage

- A. Safety Handling precaution:**
Safety Handling precaution: Ventilation (low level) is recommended when using large volumes
Use of dispensing equipment is recommended to minimise the risk of skin or eye contact
Avoid skin and eye contact.
See advice in section 8
- B. Suitable storage conditions:**
Suitable storage conditions: Storage at 2 to 8°C is recommended.

8. Exposure controls / personal protection**A. Component exposure limits:**

Hazardous components	National standard	OSHA	ACGIH
Ethyl 2-cyanoacrylate 7085-85-0	none	None	1 ppm TWA 0.2 ppm TWA

B. Engineering controls: Ensure good ventilation/extraction.

C. Personal protective equipments:

- **Respiratory protection:** In case of insufficient ventilation, wear suitable respiratory equipment.
 - **Eye protection:** Safety goggles or safety glasses with side shields.
 - **Skin protection:** Use protective gloves.
 - **Personal protection:** Do not eat, drink or smoke while working.
- Good industrial hygiene practices should be observed.

9. Physical and chemical properties

A. Appearance (physical state, color):	liquid colourless
B. Odor:	irritating
C. Odor threshold value:	Not available.
D. pH:	Not applicable, Product reacts with water.
E. Melting point / Freezing point:	Not available.
F. Boiling point:	> 149 °C (> 300.2 °F)
G. Flash point:	80 - 93 °C (176 - 199.4 °F)
H. Speed of volatilization:	Not available.
I. Flammability:	Not applicable
J. Upper flammable /Lower flammable:	Not available.
K. Vapor pressure:	< 0.2 mm hg< 26.7 Pa< 700 mbar
L. Solubility:	Polymerises in presence of water.
M. Vapor density:	Not available.
N. Specific Gravity:	Not available.
O. N-Octanol / Water partition coefficient:	Not available.
P. Self ignition point:	Not available.
Q. Decomposition:	Not available.
R. Viscosity:	18,000.00 - 40,000.00 mPa.s
S. Molecular Weight:	Not available.

10. Stability and reactivity

A. Chemical stability:	Stable under recommended storage conditions.
B. Possibility of hazardous polymerization:	Rapid exothermic polymerization will occur in the presence of water, amines, alkalis and alcohols.
C. Avoid condition (discharge of static electricity, shock, vibration):	Stable under normal conditions of storage and use.
D. Avoid materials:	Rapid exothermic polymerization will occur in the presence of water, amines, alkalis and alcohols.

E. **Decomposition products:** None if used for intended purpose.

11. Toxicological information

A. **Information for exposure route:** Skin, Inhalation, Eyes

B. **Information for health and hazard identification:**

Acute Toxicity Estimate of Mixture (ATE mix):

Oral toxicity:Not available.

Inhalative toxicity:Not available.

Dermal toxicity:Not available.

Acute toxicity:

Hazardous components CAS-No.	Value type	Value	Route of application	Exposure time	Species	Method
Ethyl 2-cyanoacrylate 7085-85-0	LD50 LD50	> 5,000 mg/kg > 2,000 mg/kg	oral dermal		rat rabbit	equivalent or similar to OECD Guideline 423 (Acute Oral toxicity) equivalent or similar to OECD Guideline 402 (Acute Dermal Toxicity)

Skin corrosion/irritation:

Hazardous components CAS-No.	Result	Exposure time	Species	Method
Ethyl 2-cyanoacrylate 7085-85-0	slightly irritating	24 h	rabbit	equivalent or similar to OECD Guideline 404 (Acute Dermal Irritation / Corrosion)

Serious eye damage/irritation:

Hazardous components CAS-No.	Result	Exposure time	Species	Method
Ethyl 2-cyanoacrylate 7085-85-0	irritating		rabbit	equivalent or similar to OECD Guideline 405 (Acute Eye Irritation / Corrosion)

Respiratory or skin sensitization:

Hazardous components CAS-No.	Result	Test type	Species	Method
Ethyl 2-cyanoacrylate 7085-85-0	not sensitising	Skin sensitisation	guinea pig	not specified

Germ cell mutagenicity:

Hazardous components CAS-No.	Result	Type of study / Route of administration	Metabolic activation / Exposure time	Species	Method
Ethyl 2-cyanoacrylate 7085-85-0	negative negative negative	bacterial reverse mutation assay (e.g Ames test) in vitro mammalian chromosome aberration test mammalian cell gene mutation assay	with and without with and without with and without		equivalent or similar to OECD Guideline 471 (Bacterial Reverse Mutation Assay) OECD Guideline 473 (In vitro Mammalian Chromosome Aberration Test) OECD Guideline 476 (In vitro Mammalian Cell Gene Mutation Test)

Carcinogenicity: No data available.

Reproductive toxicity: No data available.

Specific Target Organ Toxicity - Single exposure :

Hazardous components CAS-No.	Hazard category	Target Organs
Ethyl 2-cyanoacrylate 7085-85-0	Category 3	No data available.

Specific Target Organ Toxicity – Repeated exposure : No data available.

Aspiration hazard: No data available.

Additional Health Hazard Information

Ingredients	Hazard class	Hazard category	Route of exposure	Target organ(s)
Ethyl 2-cyanoacrylate	Skin corrosion/irritation	Category 2		
	Serious eye damage/eye irritation	Category 2		
	Target Organ Systemic Toxicant - Single exposure	Category 3		respiratory tract irritation

12. Ecological information

A. Ecological toxicity

No data available.

B. Persistence / degradability:

Hazardous components CAS-No.	Result	Route of application	Degradability	Method
Ethyl 2-cyanoacrylate 7085-85-0	not readily biodegradable.	aerobic	57 %	OECD Guideline 301 D (Ready Biodegradability: Closed Bottle Test)

C. Bioaccumulative Potential

No data available.

D. Soil Mobility

Hazardous components CAS-No.	LogPow	Bioconcentration factor (BCF)	Exposure time	Species	Temperature	Method
Ethyl 2-cyanoacrylate 7085-85-0	0.776				22 °C	EU Method A.8 (Partition Coefficient)

E. Other Adverse Effects: Do not empty into drains, soil or bodies of water.

13. Disposal considerations

A. Disposal method:

Dispose of in accordance with local and national regulations.

B. Waste information (including waste method of contaminated container and packaging):

Waste disp. packag. not clean: Disposal must be made according to official regulations.

14. Transport information

Road transport ADR:

Not dangerous goods

Railroad transport RID:

Not dangerous goods

Inland water transport ADN:

Not dangerous goods

Marine transport IMDG:

Not dangerous goods

Air transport IATA:

A. UN number:	3334
B. UN proper shipping name:	Aviation regulated liquid, n.o.s. (Cyanoacrylate ester)
C. Transport hazard class:	9
D. Packing group (if applicable):	III
E. Marine pollution (yes/no):	
F. Special precaution which a user to be aware of or needs to comply with in connection with transport or conveyance either within or outside their premises:	No data available.
Packaging instructions (passenger):	964
Packaging instructions (cargo):	964
Label:	9
Additional Information:	Primary packs containing less than 500ml are unregulated by this mode of transport and may be shipped unrestricted.

15. Regulatory information

A. According to Industrial Safety & Health Act:

Harmful Substances Prohibited from Manufacturing:

Neither banned nor restricted

Harmful Substances Requiring Permission for Manufacture or Use:

Neither banned nor restricted

Harmful Substances Requiring Workplace Environment Monitoring:

Neither banned nor restricted

Controlled Hazardous Substances:

Neither banned nor restricted

Harmful Substances Requiring Special Medical Examination:

Neither banned nor restricted

Korea OELs:

Neither banned nor restricted

B. According to Chemicals Control Law:

Toxic Chemicals:

Neither banned nor restricted

Banned Toxic Chemicals:

Neither banned nor restricted

Restricted Toxic Chemicals:

Neither banned nor restricted

Accidental Release Prevention Substances:

Neither banned nor restricted

C. According to Dangerous Substances Safety Management Act Enforcement Rule :
4th Class Flammable Liquids, Class 3 Petroleum, Hazard Class III

D. According to Enforcement Decree of The wastes control Act :
Designated Wastes, Appendix 4, Code Numbers.
Designated Wastes

E. According to other regulations:
Not available.

16. Other information

- A. Reference:** msds.kosha.or.kr/MSDSInfo
IUCLID
Henkel MSDSetc.
NCIS
- B. Date of creation:** 03.06.2022
- C. Revision number and the latest version date** V001.0
03.06.2022
- D. Disclaimer:** This Safety Data Sheet has been generated based on KOSHA Standard (MoEL Notice No. 2020-130) and provides information in accordance with Korea law only. No warranty or representation of any kind is given with respect to the substantive or export laws of any other jurisdiction or country. Please confirm that the information provided herein conforms to the substantive export or other law of any other jurisdiction prior to export. Please contact Henkel Product Safety and Regulatory Affairs for additional assistance. This information is based on our current level of knowledge and relates to the product in the state in which it is delivered. It is intended to describe our products from the point of view of safety requirements and is not intended to guarantee any particular properties. The data contained herein are furnished for information only and are believed to be reliable. However, Henkel Corporation and its affiliates ("Henkel") does not assume responsibility for any results obtained by persons over whose methods Henkel has no control. It is the user's responsibility to determine the suitability of Henkel's products or any production methods mentioned herein for a particular purpose, and to adopt such precautions as may be advisable for the protection of property and persons against any hazards that may be involved in the handling and use of any Henkel's products. In light of the foregoing, Henkel specifically disclaims all warranties, express or implied, including warranties of merchantability and fitness for a particular purpose, arising from sale or use of Henkel's products. Henkel further disclaims any liability for consequential or incidental damages of any kind, including lost profits.